Long Research Stay:

Swiss Institute of Bioinformatics (SIB)

Elias Crum

KNoWS Presentation
May 15th, 2025

Talk Outline

fwo Long Research Stay Grant Application

Logistics and Takeaways

Highlights of Research Stay

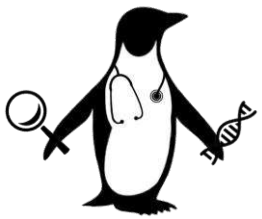
SIB Offices Visited

Research worked on

Planned Publications / Collaborations

Future aims

PENGQUIN: PErsoNal Genome QUery IN healthcare and clinical practice



Genomic Data → Clinical Care

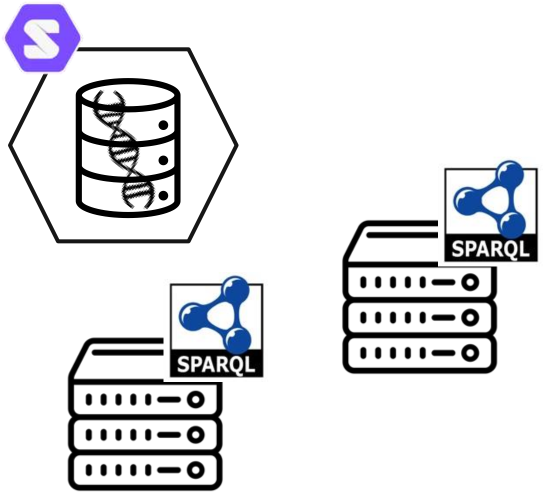
- Semantic, Linked Data
- Query Capabilities
- Framework + App

Advisors: Dr. Gökhan Ertaylan[1], Dr. Bart Buelens[1], Dr. Ruben Taelman[2], Prof. Dr. Ruben Verborgh[2]

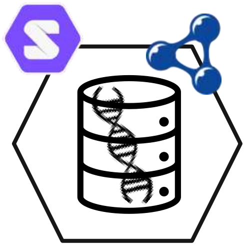
1 = vito 2 = GHENT UNIVERSITY

PENGQUIN Goals

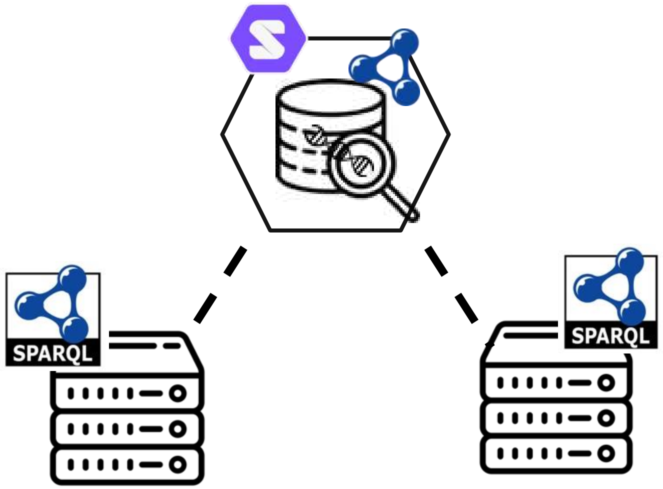
Data Storage



Genomic Data Representation



Question Answering



Framework
& web application

GENOME Pods

Research Stay Motivation

The Swiss Institute of Bioinformatics (*SIB*)

↳ Research Organization (not an Academic Institution)

- Technically my research stay was an “internship”

↳ SIB Mission

- Progress data science through in-depth knowledge of biological data, cutting-edge technologies, and interdisciplinary collaborations

Why a research stay here?

↳ Large, Public SPARQL Endpoints

↳ Swiss Personalized Health Network (SPHN)

↳ Pre-existing Collaboration – CHIST-ERA TRIPLE Project

Funding – FWO Long Research Stay Grant

Long Research Stay Application

↳ Submission due 3-months prior to stay

↳ Application Materials

- Written Application (max 5000 chars each)
    1. What is the added scientific value of the stay?
    2. Motivation for the choice of the non-Belgian host organization
        - Including 3 publications from the organization
    3. Provide a work plan, i.e. the different work packages and a detailed timetable

- 3 Personal Publications

- Letter of Recommendation from your Promoter

- Letter Invitation from host organization

Grant Application Takeaways

Hardest part is coordinating with host organization

Overall, was quite low intensity

Monetary award is suitable for ~most~ places (Switzerland was a bit of a stretch…)

I would HIGHLY recommend taking advantage of this

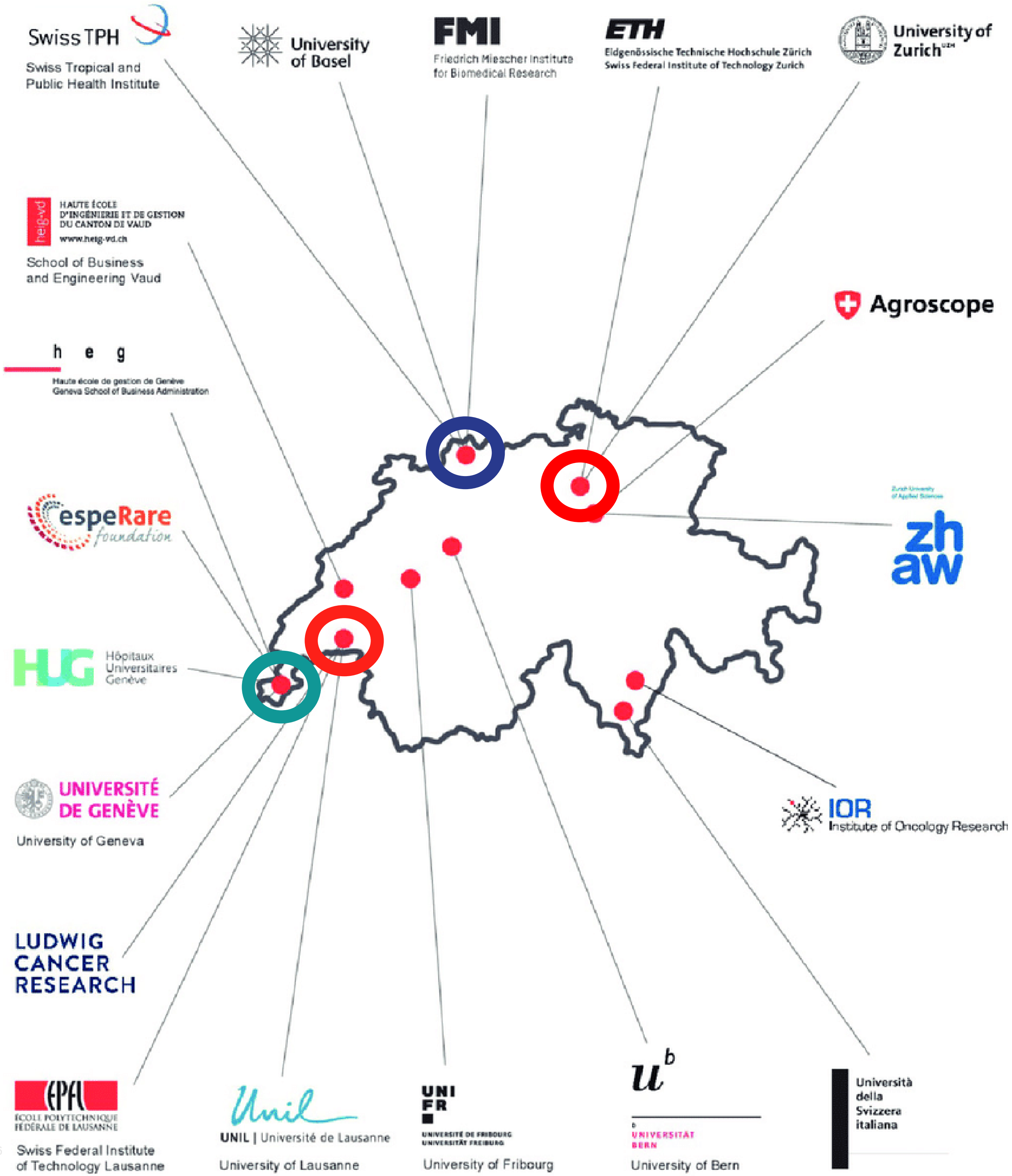
The Research Stay

A disappointing start …

↳ Arrived in Switzerland to discover my accommodations were fraudulent (big sad face)

↳ After that, it was very enjoyable!

The SIB Network



SwissProt

Swiss Personalized Health Network (SPHN)

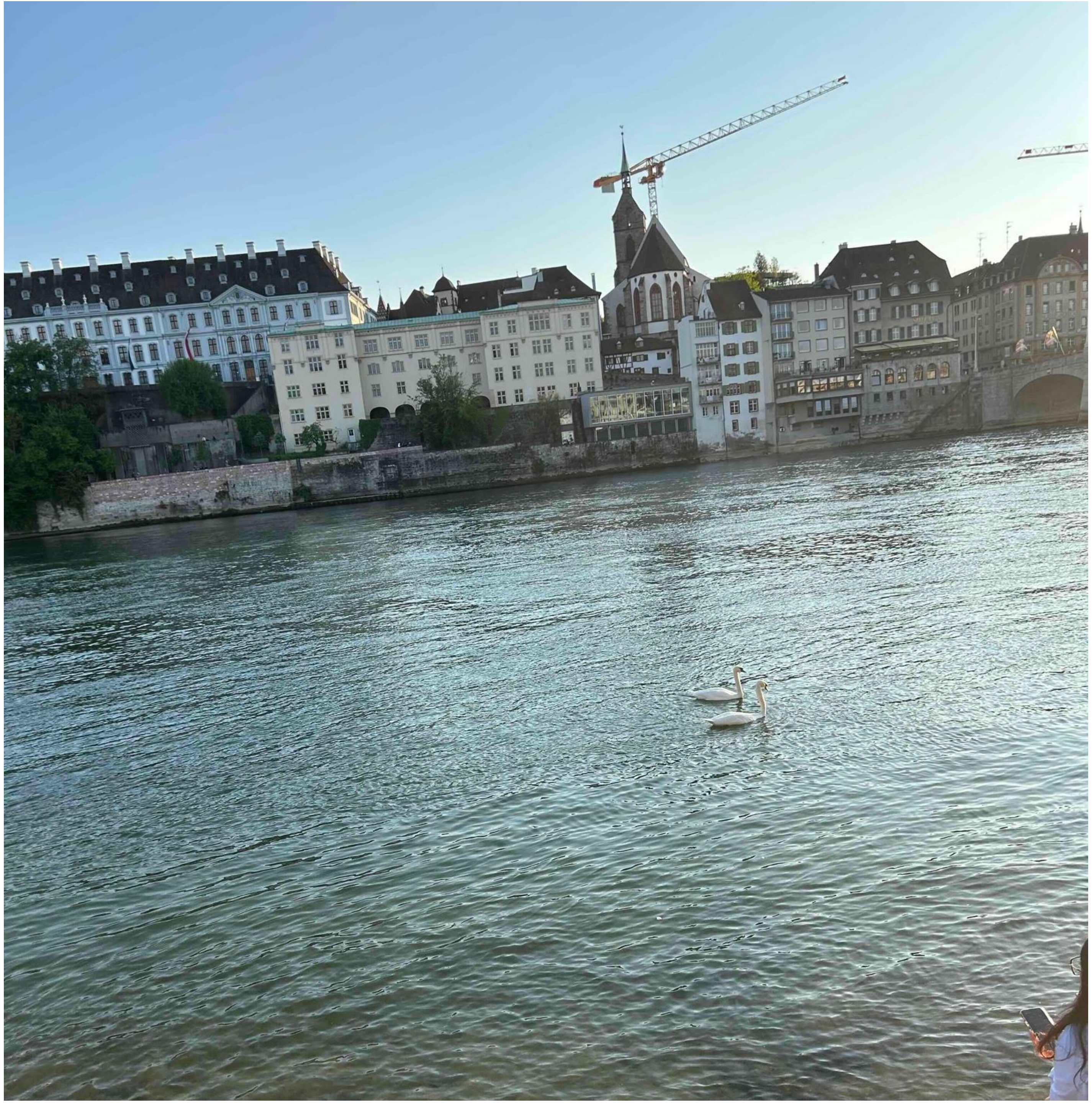
SIB Vital-IT


GERMANY
FRANCE
Basel
Zurich
Sankt Gallen
AUSTRIA
Zug
Bern
Neuchatel
Fribourg
Chur
SWITZERLAND
Lausanne
Sion
Geneva
ITALY

Lausanne

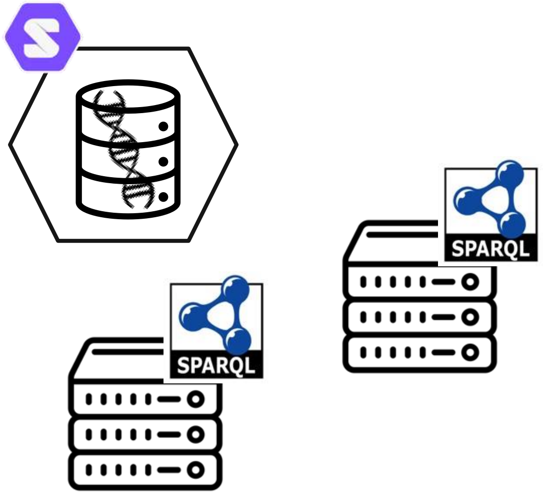
Zurich

Basel

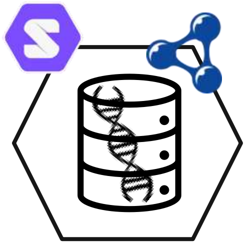


What I Worked On:

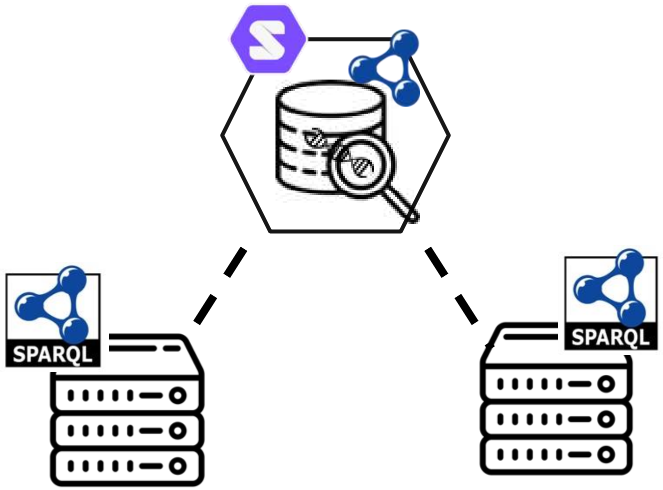
Data Storage



Genomic Data Representation

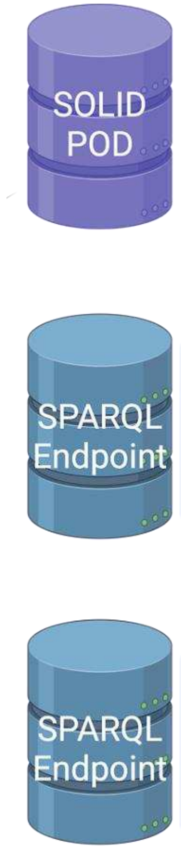


Federated Querying



CHIST-ERA TRIPLE Project

Swiss Institute of Bioinformatics
GHENT UNIVERSITY
ÚOCHB IOCB PRAGUE
chist-era



Standardized documentation and resource statistics

Data Storage Progress

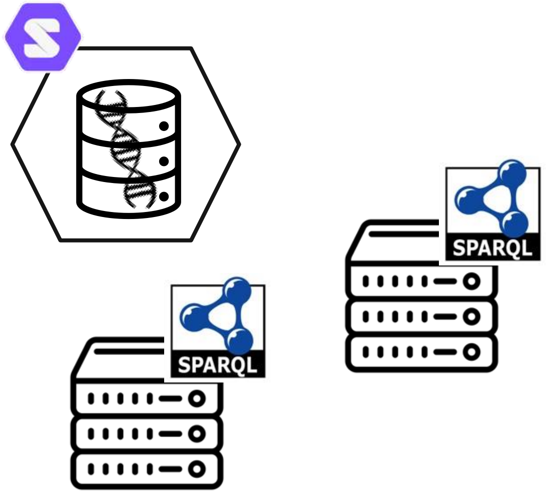
A Web Application to:

knowledgeonwebscale.github.io/solid-cockpit/

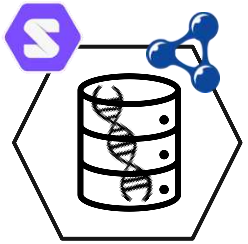
- *Upload Data to a Solid Pod*
- *Browse contents of a Solid Pod*
- *Edit Privacy of (and share) data in a Solid Pod*
- *Query Data in Solid Pods + SPARQL Endpoints*
  - ↳ *Implement a query cache for storing queries and results*

Semantifying Genomic Variant Data

Data Storage

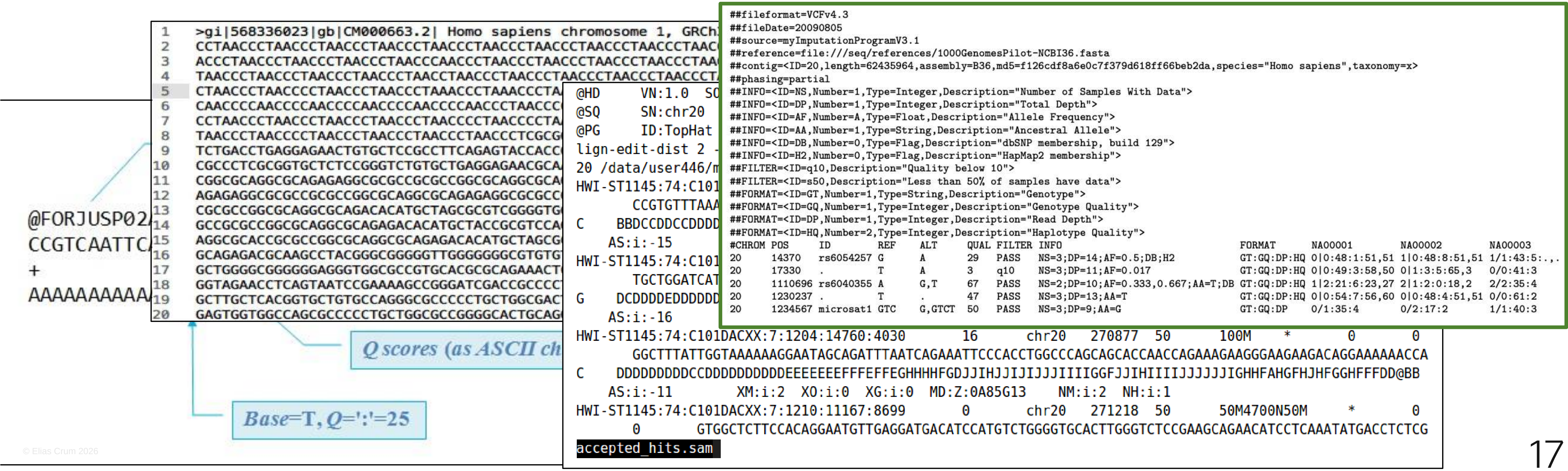


Genomic Data Representation



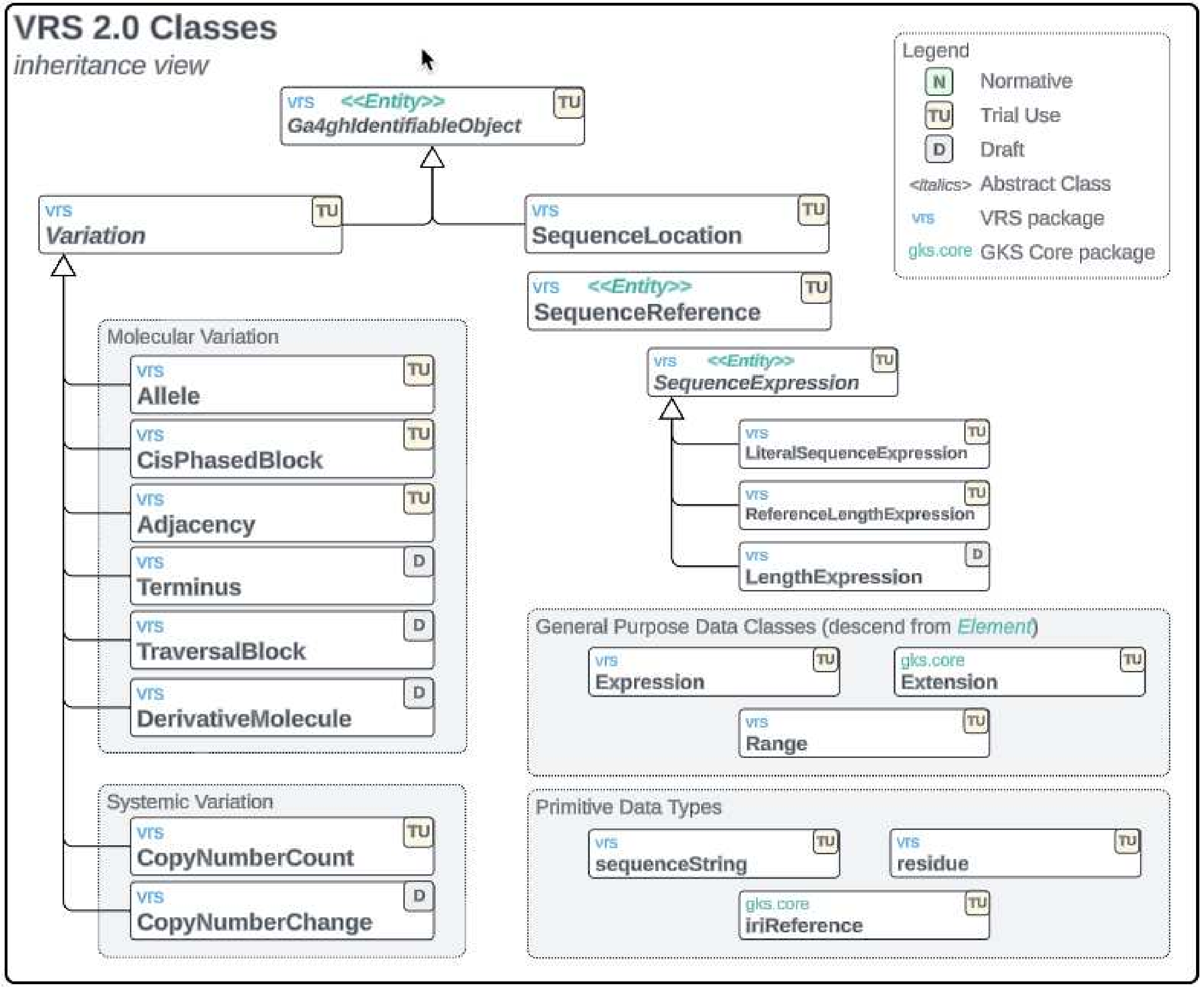
Semantic VCF – *Sequencing Data Formats Intro*

FASTQ	+	FASTA	→	SAM/BAM	→	VCF
Raw Sequencing Outputs		Contiguous DNA sequence		FASTQ reads aligned to a FASTA reference		Areas in SAM where sample and reference differ



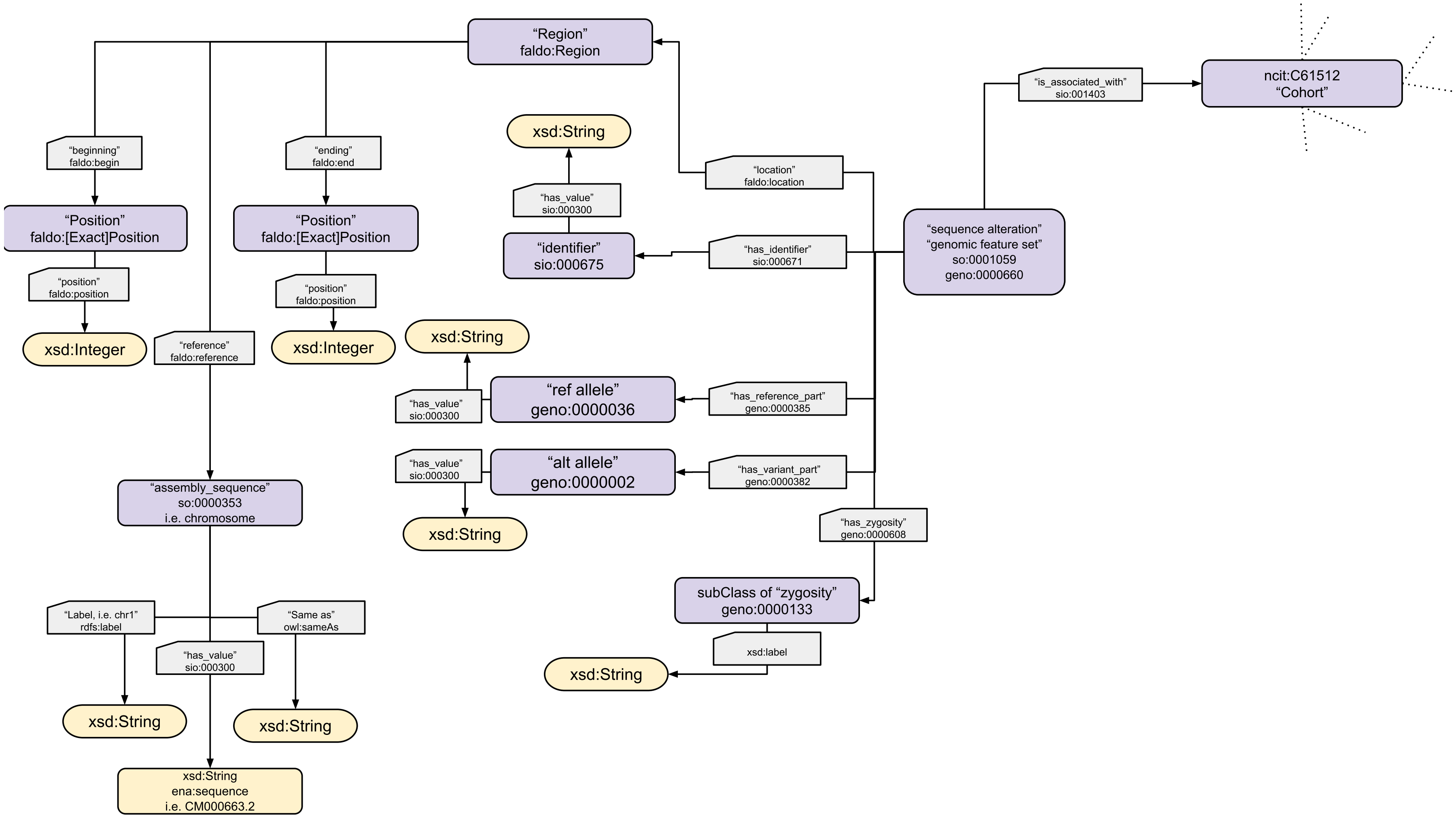
Representing VCF Semantically?

GA4GH Variant Schema:



Existing Semantic VCF Resources – Semantic Beacons

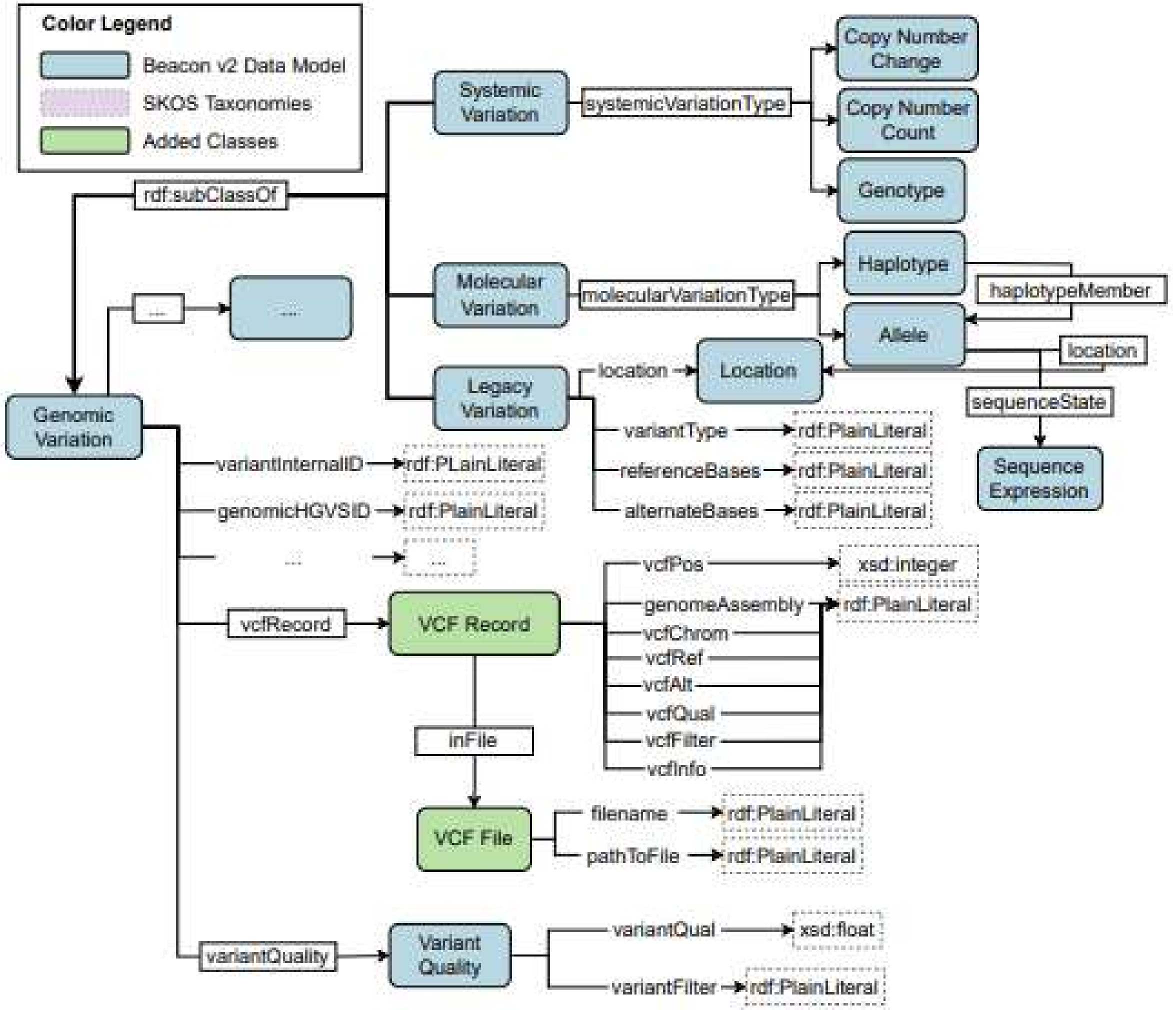
Semantic Beacons Schema:



https://gitlab.univ-nantes.fr/bodrug-a/genomic-variants-schema

Existing Semantic VCF Resources – HERO Genomics

HERO Genomics Ontology:



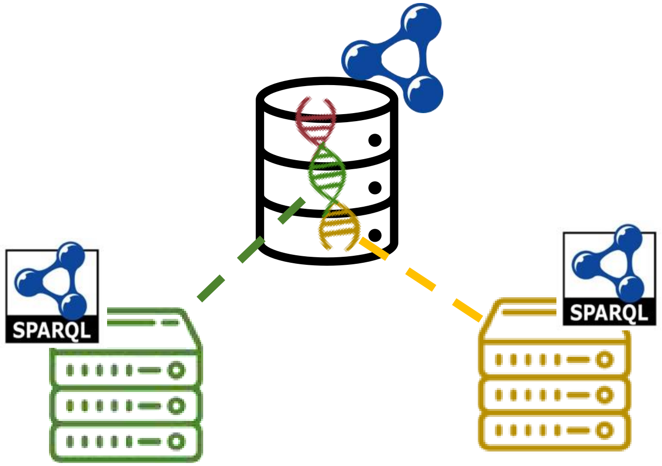
https://www.dei.unipd.it/~silvello/papers/2025_MCS_SWAT4HCLS.pdf

Semantic VCF - Future Aims

Construct VCF Knowledge Graph

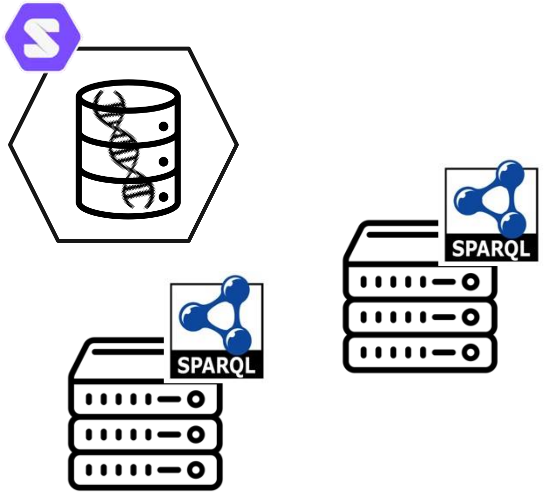
manual conversion **OR** RocketRML

Experiment with SPARQL Querying and Data Linking

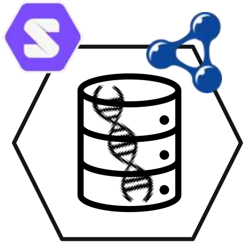


SPARQL Federated Querying

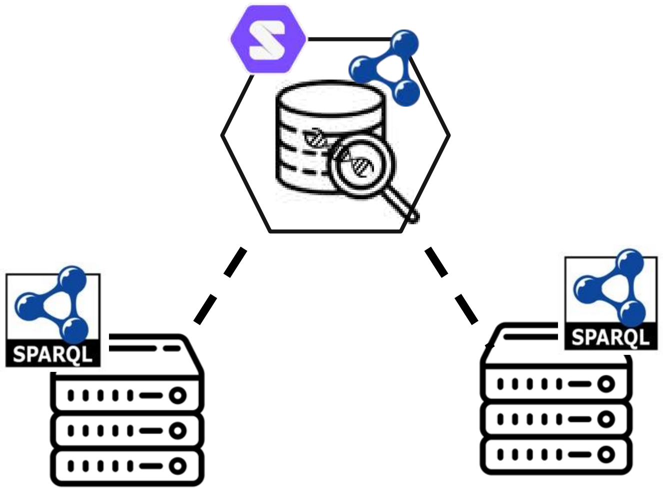
Data Storage



Genomic Data Representation



Querying



Real World Federated Query Survey

Swiss Institute of Bioinformatics
GHENT UNIVERSITY

Motivation:

Federated querying for answering questions over **multiple heterogeneous data sources**

Performance is currently ~*variable*~

We want to detail *WHY* performance varies so much && suggestions for improvement

Assess federation algorithms on REAL endpoints

FedX – uses `ASK` queries to inform source-selection

SPLENDID – `VoID` descriptions to inform source-selection

Real World Federated Query Survey *- Design*

SPARQL Endpoints:

"Real World" Federated Queries: https://sib-swiss.github.io/sparql-examples/

SPARQL Query Engine: https://comunica.dev/

Example Federated Query

Query: 027-biosodafrontend

Question: (*Bgee* | *UniProt*)

Retrieve the *genes* expressed in the *human pancreas*

and

those *genes' annotations*

https://www.bgee.org/sparql/

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX up: <http://purl.uniprot.org/core/>
PREFIX obo: <http://purl.obolibrary.org/obo/>
PREFIX genex: <http://purl.org/genex#>
PREFIX lscr: <http://purl.org/lscr#>
PREFIX orth: <http://purl.org/net/orth#>
SELECT DISTINCT ?geneEns ?uniprot ?annotation_text {
    {
        SELECT ?geneEns {
            ?geneB a orth:Gene .
            ?geneB genex:isExpressedIn ?cond .
            ?cond genex:hasAnatomicalEntity ?anat .
            ?geneB lscr:xrefEnsemblGene ?geneEns .
            ?anat rdfs:label 'pancreas' .
            ?geneB orth:organism ?o .
            ?o obo:RO_0002162 ?taxon2 .
            ?taxon2 up:commonName 'human' .
        } LIMIT 100
    }
    SERVICE <https://sparql.uniprot.org/sparql> {
        ?uniprot rdfs:seeAlso / up:transcribedFrom ?geneEns .
        ?uniprot up:annotation ?annotation .
        ?annotation rdfs:comment ?annotation_text .
    }
}
```

Example obtained from: https://biosoda.expasy.org/build_biosodafrontend/

visualization from: https://sib-swiss.github.io/sparql-editor/

Real World Federated Query Survey – *[Preliminary] Findings 1*

Preliminary Results (endpoints and queries):

Federated querying is not ”easy”

↳ Errors related to SPARQL endpoint availability

Invalid SPARQL endpoint response from https://semantic.eea.europa.eu/sparql (HTTP status 404)

↳ Errors caused by SPARQL endpoint-related errors

Invalid SPARQL endpoint response from https://sparql.orthodb.org/sparql (HTTP status 500)

↳ Errors related to SPARQL endpoint rate-limiting

Invalid SPARQL endpoint response from https://query.wikidata.org/sparql (HTTP status 429): Rate limit exceeded

↳ Errors related to SPARQL endpoint time-outs

Unexpected ""!"" at position 0 in state STOP

Real World Federated Query Survey – *[Preliminary] Findings 2*

Preliminary Results (algorithms):

Federated querying is not "easy"

↳ Errors within Comunica

Actor urn:comunica:default:query-operation/actors#source requires an operation with source annotation.

fetch failed

Could not retrieve https://idsm.elixir-czech.cz/sparql/endpoint/chebi (HTTP status 400)

FATAL ERROR: Ineffective mark-compacts near heap limit Allocation failed - JavaScript heap out of memory

↳ Errors related to algorithms

Actor urn:comunica:default:/actors#rdf-join cannot complete join with Asym-hash actor.

Planned Publications

Real-World Federated Query Survey (conference)

↳ SEMANTiCS 2025 (Oct 2025)

- Long Research Paper (May 23rd)
- Get feedback from community

Real-World Federated Query Survey (journal)

↳ Journal Undecided (~Winter 2025-2026)

Solid Cockpit (conference)

↳ ESWC 2026 (June 2026)

- Resource Paper (Dec 2025)

VCF to RDF Implementation (conference)

↳ SWAT4HCLS 2026 (Feb 2026)

- Full Paper (Dec 2025)

Planned Collaborations

Real-World Fed Query Survey

- 3 contributing authors from SIB
- Planning to present findings in Switzerland after submissions
- CHIST-ERA Project finishes in Oct 2025
- UGent promoters submitted joint grant proposal for 2026-2030

VCF to RDF Follow-Up

- Planned extension to existing SHPN RDF Schema
- Will keep in close contact

SIB Research Stay: 1 March – 30 April

Context + Thank You!

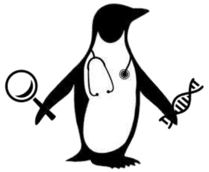
Tarcisio Mendes

Ana-Claudia Sima

Jerven Bolleman

Marco Pagni

Thank you!



More PhD specifics

LinkedIn

GitHub

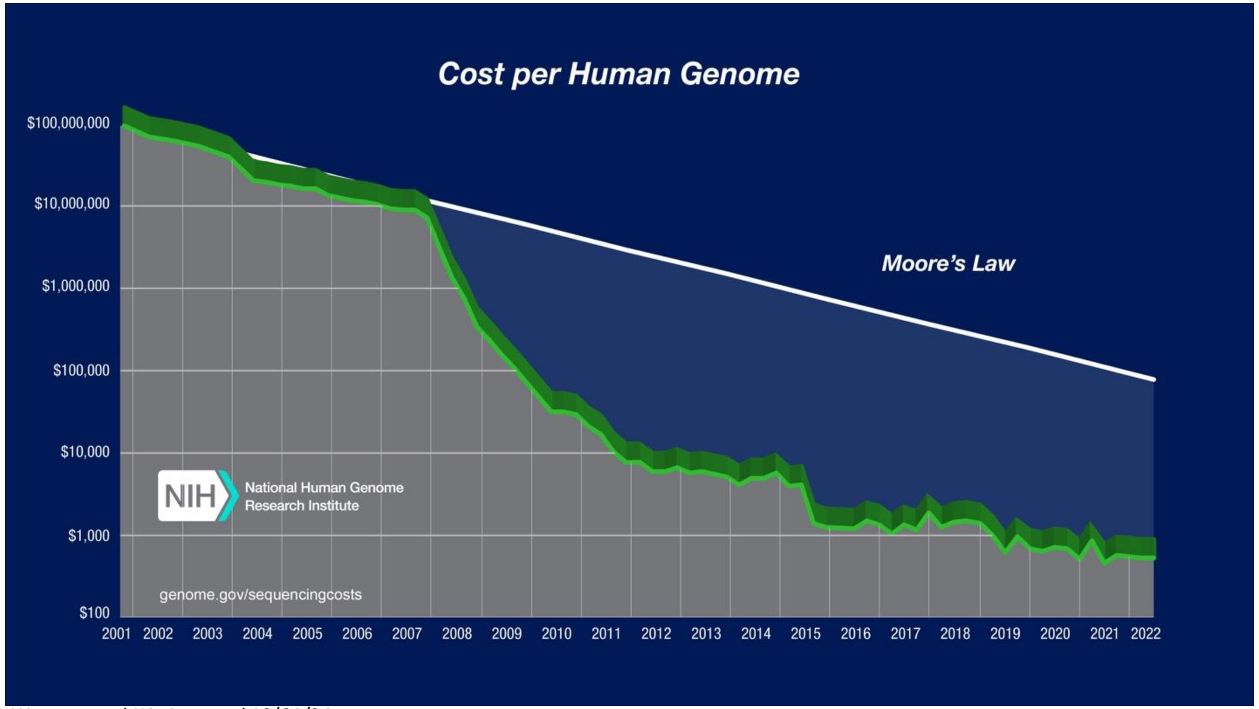
Supplemental Slides

Solid Cockpit

knowledgeonwebscale.github.io/solid-cockpit/

Genome Sequencing in Health Care

Sequence Generation Costs



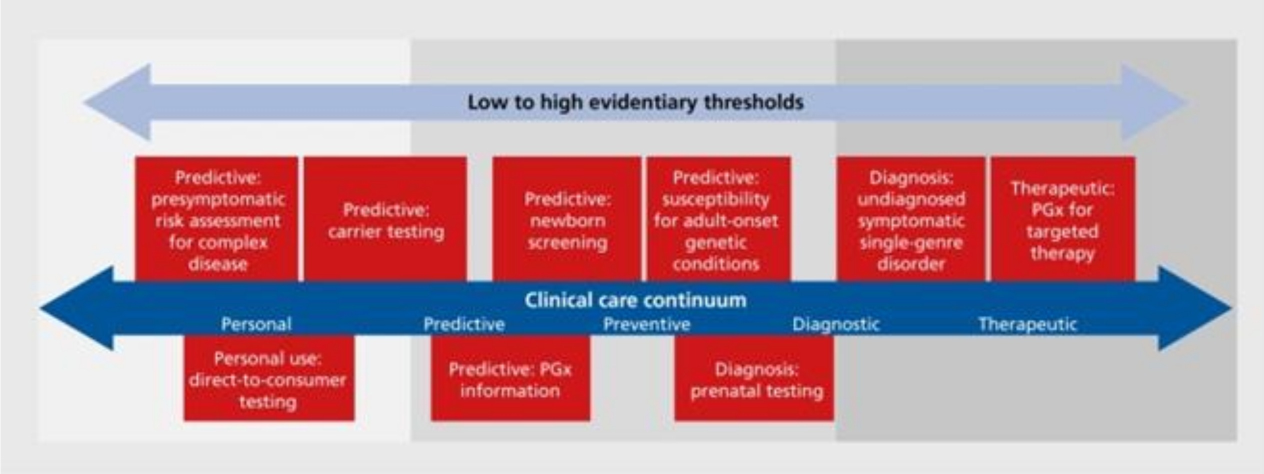
Wetterstrand KA. Accessed 12/04/24.

Currently getting close to $100 per genome[4]

Per-base cost of sequencing has been dropping by half every ~5 months

Per-base cost of sequence data storage is dropping by half every ~14 months

Use-cases



10.31887/DCNS.2016.18.3/jkrier

Examples include:

- Medication prescription (pharmacogenomics)
- Genetic disease screening
- Cancer diagnosis & treatment

Variant Call Format (VCF)

- tab delimited .txt file

- Rows represent individual mutations in a sequence (when compared to reference)

- Columns store information about that mutation

#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT

```
##fileformat=VCFv4.1
##fileDate=20140930
##source=23andme2vcf.pl https://github.com/arrogantrobot/23
##reference=file://23andme_v3_hg19_ref.txt.gz
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM  POS ID  REF ALT QUAL    FILTER  INFO    FORMAT  GEN
chr1    82154   rs4477212   a   .   .   .   .   GT  0/0
chr1    752566  rs3094315   g   A   .   .   .   GT  1/1
chr1    752721  rs3131972   A   G   .   .   .   GT  1/1
chr1    798959  rs11240777  g   .   .   .   .   GT  0/0
chr1    800007  rs6681049   T   C   .   .   .   GT  1/1
chr1    838555  rs4970383   c   .   .   .   .   GT  0/0
chr1    846808  rs4475691   C   .   .   .   .   GT  0/0
chr1    854250  rs7537756   A   .   .   .   .   GT  0/0
chr1    861808  rs13302982  A   G   .   .   .   GT  1/1
chr1    873558  rs1110052   G   T   .   .   .   GT  1/1
chr1    882033  rs2272756   G   A   .   .   .   GT  0/1
chr1    888659  rs3748597   T   C   .   .   .   GT  1/1
chr1    891945  rs13303106  A   G   .   .   .   GT  0/1
```

Personal Genome Sequence Data

Human genome data is big...

~3.2 BILLION base pairs ----> ~1 GB

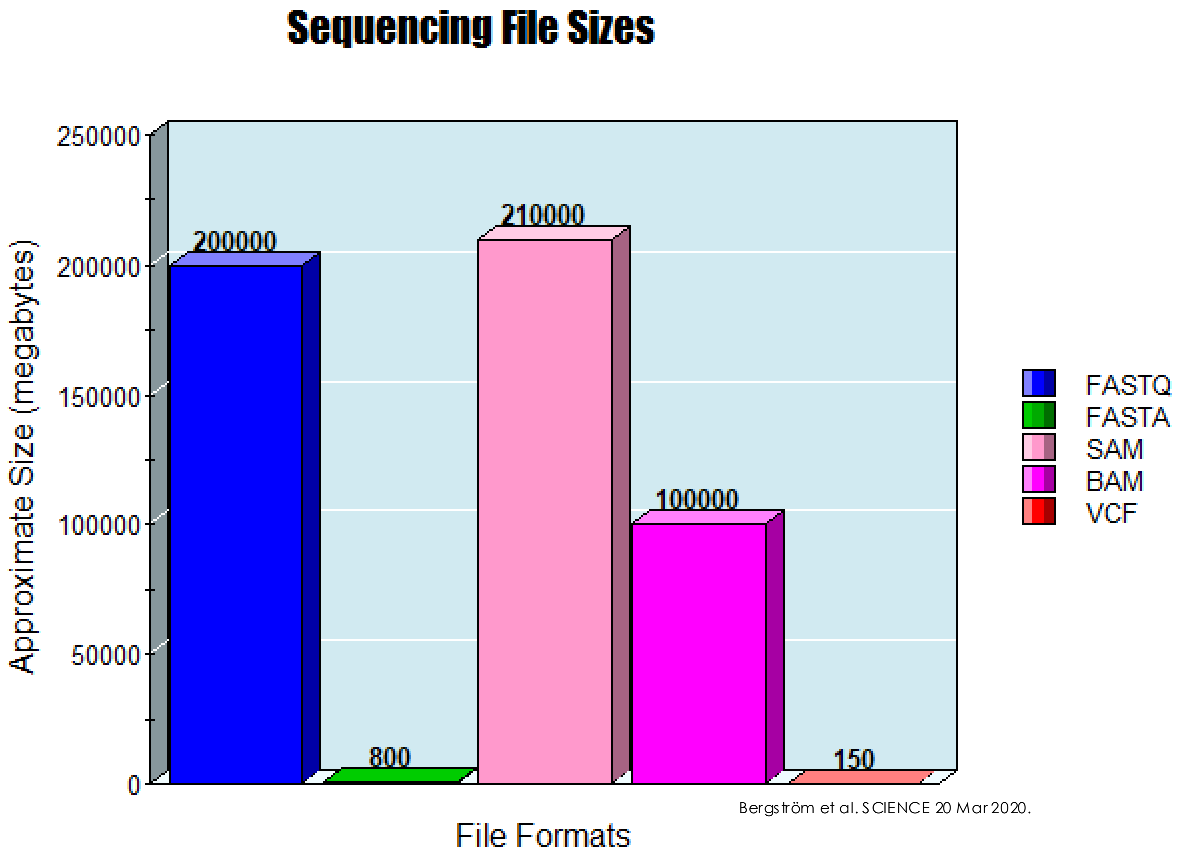
Sequencing output file ----> ~200 GB

Variant Call Format (***VCF***) file ----> ~200 MB

- Individual humans exhibit ~3 million mutations
- VCF file **records only those differences**
- Flat file

VCF file semantic representation

- **~ >27 MILLION** Triples when represented as RDF
- Ontology for conversion
- Header Dictionary Triples (for compression)



Bergström et al. SCIENCE 20 Mar 2020.

Scaling Genome Data Usage

Research genomic data sharing initiatives:

FHIR

Beacon

OHDSI
OBSERVATIONAL HEALTH DATA SCIENCES AND INFORMATICS

State-of-the-art for **clinical** genomic data sharing?

GDPR (EU) / HIPPA (US)
EU Health Data Spaces

Not much...

Experimental VCF Data

Primary dataset from publicly available repositories (>100 genomes)

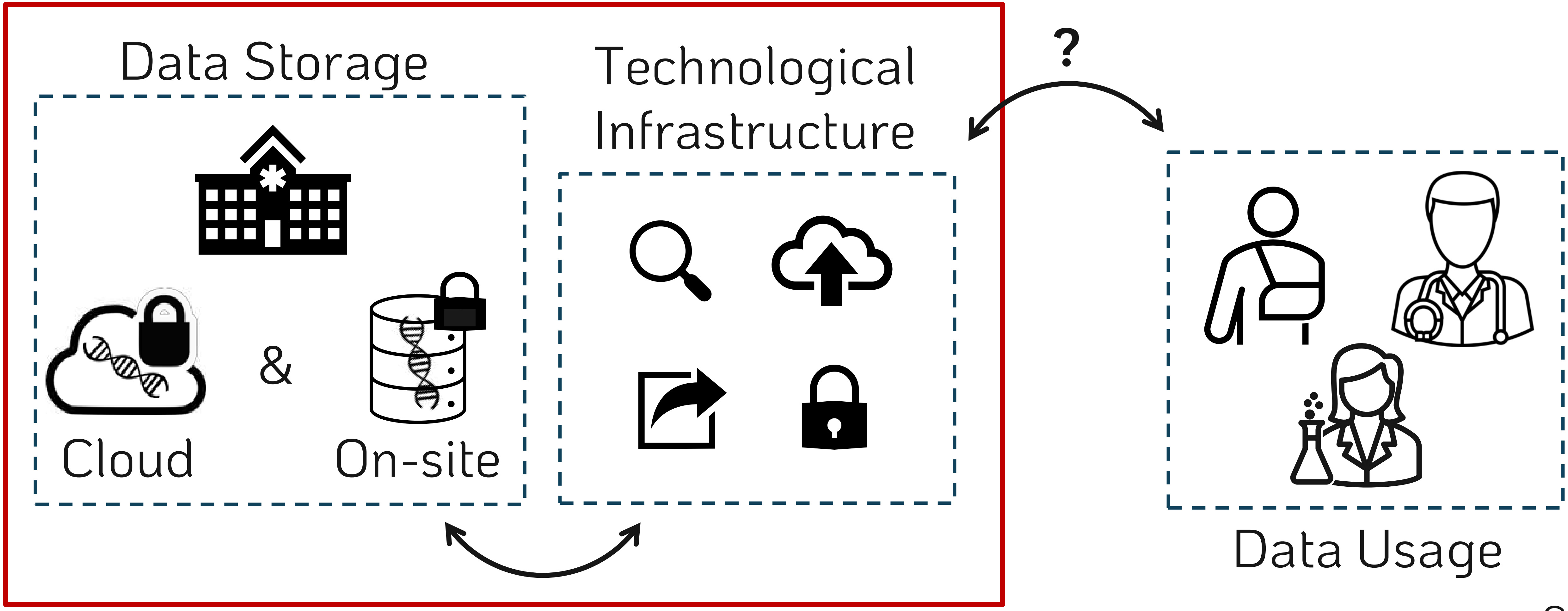
- Personal Genome Project (https://my.pgp-hms.org/public_genetic_data)
- NIST's Genome in a Bottle public dataset (https://data.nist.gov/od/id/mds2-2336)
- Illumina Platinum Genomes repository (https://emea.illumina.com/platinumgenomes.html)

If more are needed (>1000 [research grade] genomes):

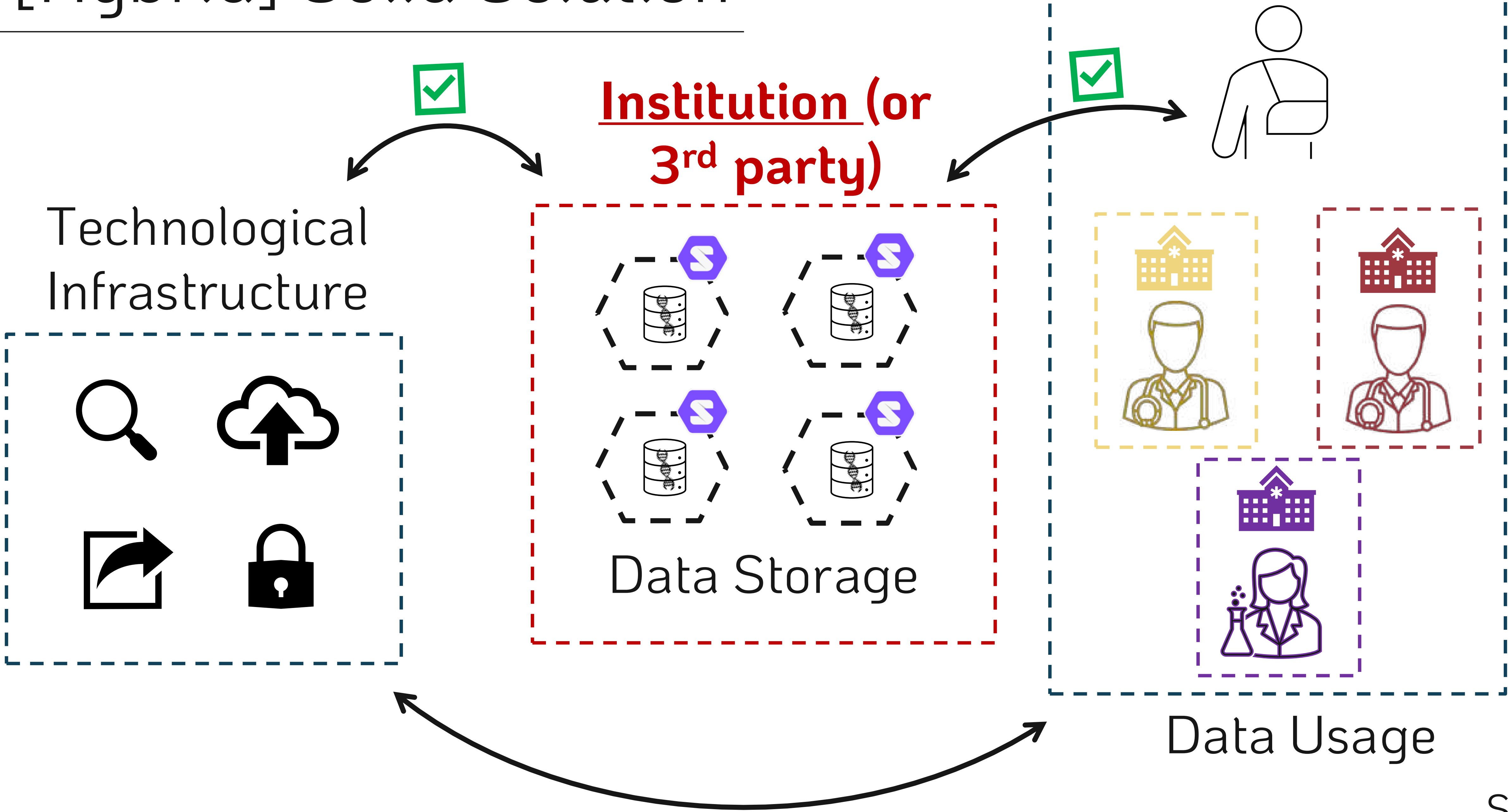
- 1000 genomes project (https://www.internationalgenome.org/)
- NCBI's ClinVar (https://www.ncbi.nlm.nih.gov/clinvar/)

Semantic Genome Data?

Institution-centric



A [Hybrid] Solid Solution



Solid Technology

- Does not solve ALL problems, but offers an environment where improvements are possible

  - ✓ Easier sharing (web-facilitated)
  - ✓ Granular privacy controls
  - ✓ Direct patient data access
  - ✓ Data representation flexibility (RDF)
  - ✓ Possibility of data-linking and querying

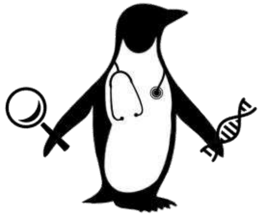
https://solidproject.org/TR/protocol

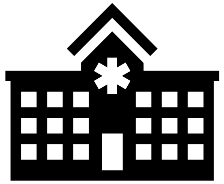
Project Goals

Create a proof-of-concept
storage **FRAMEWORK**
and accompanying **Web Application**

While **exploring representation + querying limits**

Envisioned Product Evolution





PhD End | Short Term | Medium Term | Long Term

PhD End: PhD Deliverable

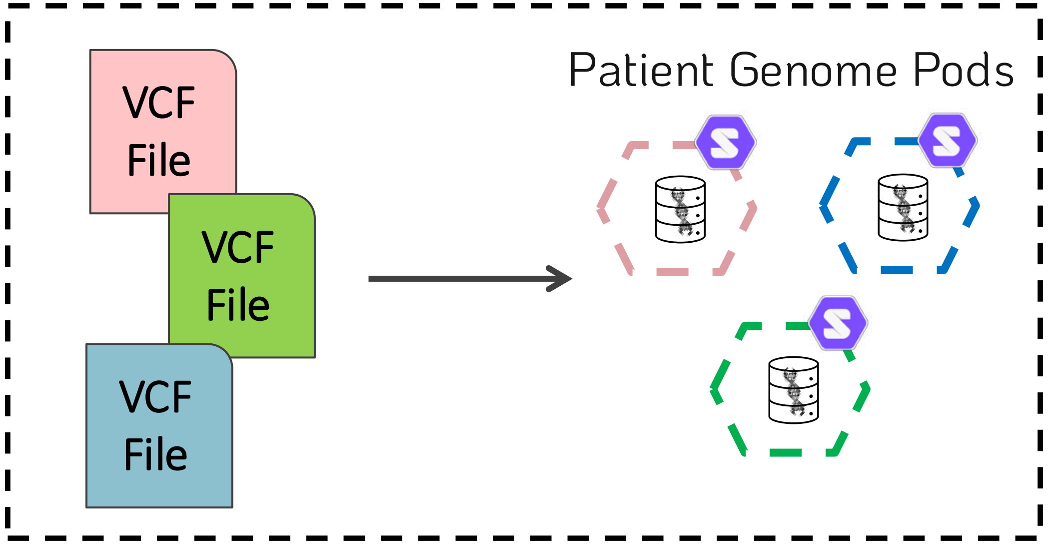
Short Term: Start-up / Product development + Hospital implementation

Medium Term: Integrate with EHR + Increase application domains

Long Term: Increase 3rd party application support + Scale to Europe and beyond

Workflow Overview

Data Storage



Data Storage

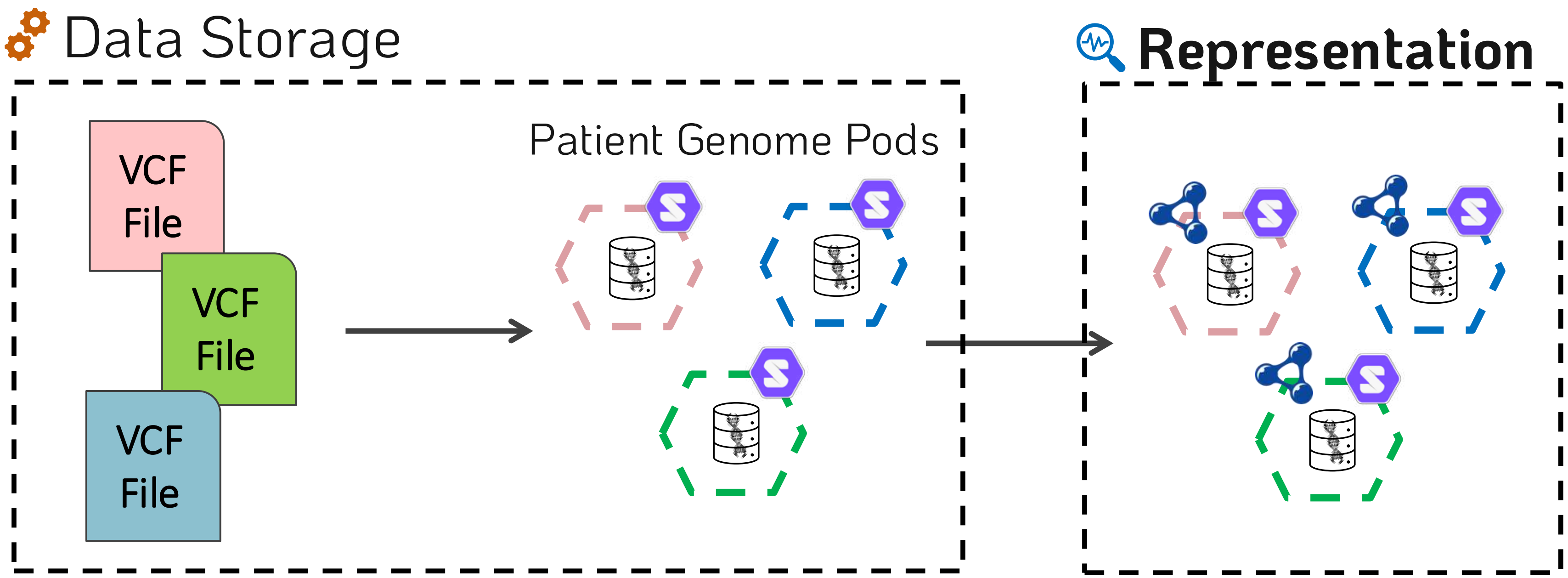
Solid can be implemented for genomic data storage.

Solid pod instances → Community Solid Server[1]

VCF Genomic Data → Illumina Platinum Genomes[2]

[1]https://github.com/CommunitySolidServer/CommunitySolidServer
[2]https://emea.illumina.com/platinumgenomes.html

Workflow Overview

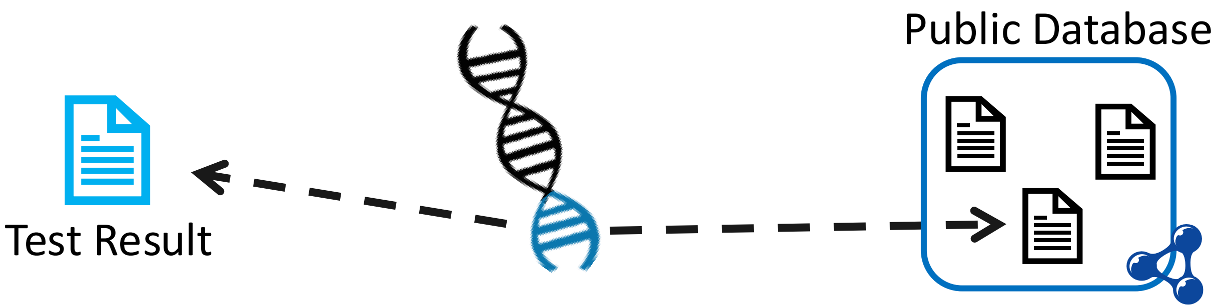


Genome Data in RDF

VCF genomic data can be represented as RDF.

VCF to RDF Vocabularies exist ⟶ SPHN Semantic Interoperability Framework[4] + FHIR HL7 Format[5]

VCF genomic data can be Linked to other data.

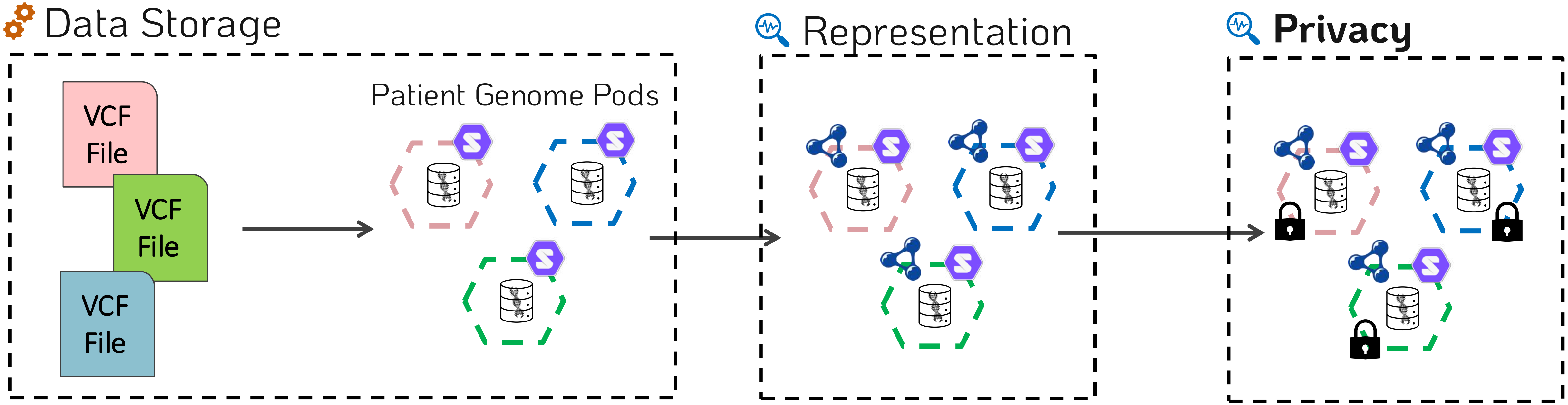


How to automate this?
What vocabularies could be used for the linkages?

[3] van der Horst E, et al. Bridging Clinical and Genomic Knowledge: An Extension of the SPHN RDF Schema for Seamless Integration and FAIRification of Omics Data. 2023.

[4] Dolin RH, et al. vcf2fhir: a utility to convert VCF files into HL7 FHIR format for genomics-EHR integration. 2021.

Workflow Overview



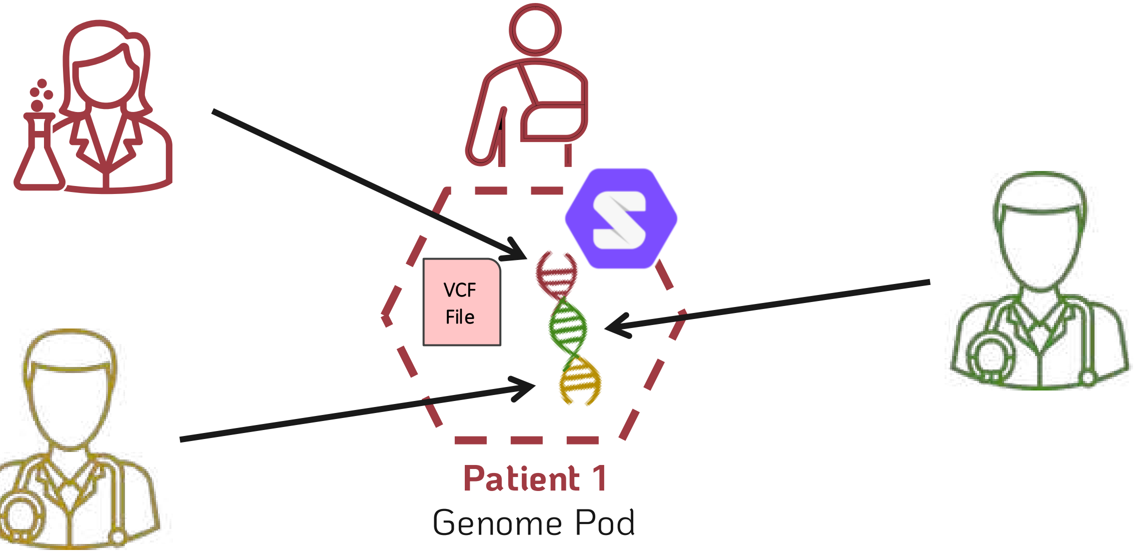
Data Privacy Controls

Granular privacy controls can be implemented in Solid for genomic data.

Web Access Contol → Solid Protocol ACP[3]

How granular can we make the privacy controls?

Privacy controls at chromosome level, or even mutation level.



Chromosome is a semantic Class

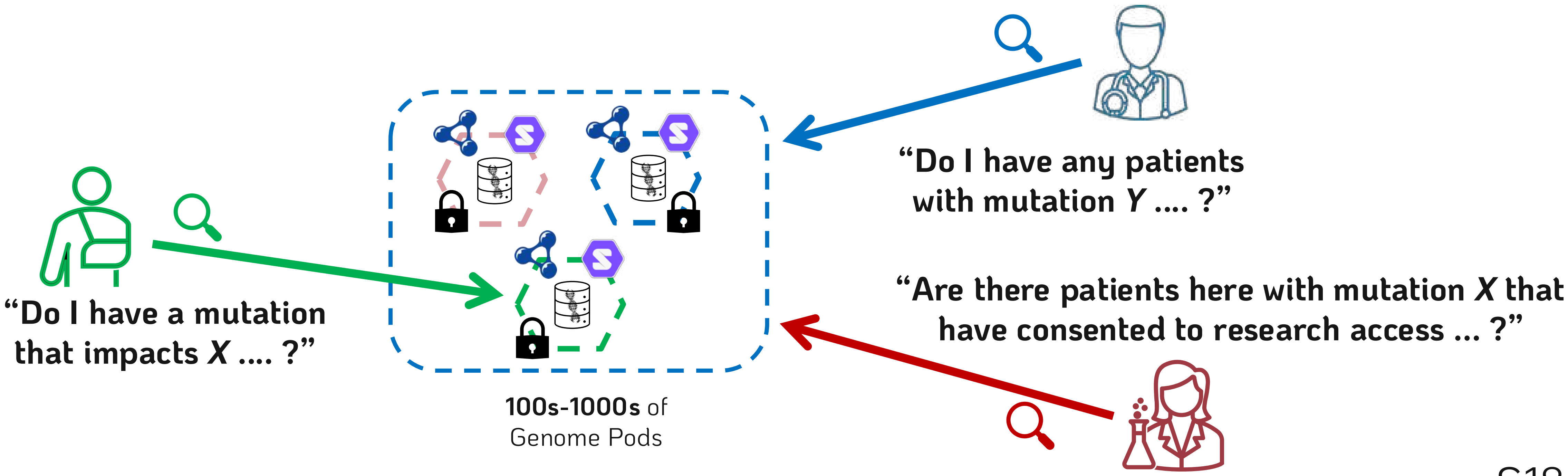
Mutation is also a semantic Class, above a triple but below a Chromosome

[5] https://solidproject.org/TR/protocol

Querying Genome Pods

RDF genomic data pods can be queried.

Query Engine → Custom Comunica implementation/engine[6]



[6] https://comunica.dev/

Querying Genome Pods (possibilities)

Single Pod Query

Utilizes an intra-pod index

Many Pod Query

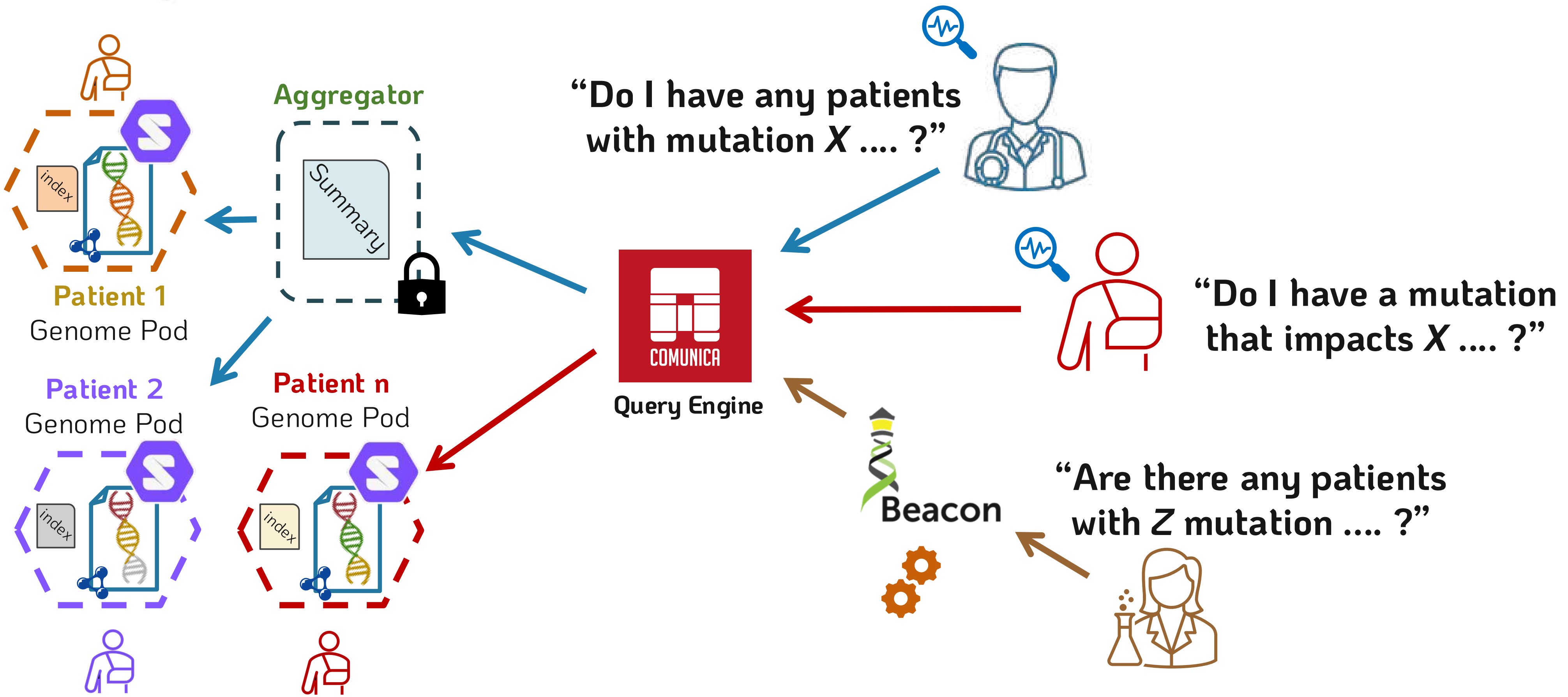
First, utilizes an extra-pod aggregator

Then, utilizes the intra-pod indexes

Beacon API for Researcher Queries

Query is translated from Beacon API to Comunica

Then, the **Many Pod** query approach is followed



Implementation Specifics

Solid pod instances → Community Solid Server[1]

Genomic Data → Publicly available *VCF* data[2]

RDF Conversion → HERO Genomics + SPHN ontology[4]

Data-linking → Actively exploring ontologies / methodologies

Data Querying → Custom instance of Comunica[6]